Recuperação Paralela - 2ª Série - 21/05

Exercício 1 - Alternativa A

$8\ mol\ H_2SO_4$ — $1L$
x — $0,4L$

$x = 3,2\ mol\ H_2SO_4$

$2\ mol\ H_2SO_4$ — $1L$
$3,2\ mol\ H_2SO_4$ — y

$y = 1,6L$ volume final

$1,2L$ de H_2O

Exercício 2 - Alternativa B

$1\ mol\ NaOH$ — $1L$
x — $4,9L$

$x = 4,9\ mol\ NaOH$

$$H_2SO_{4(aq)} + 2NaOH_{(aq)} \rightarrow Na_2SO_{4(aq)} + 2H_2O_{(l)}$$

$1\ mol\ H_2SO_4$ — $2\ mol\ NaOH$
y — $4,9\ mol\ NaOH$

$y = 2,45\ mol\ H_2SO_4$

$2,45\ mol\ H_2SO_4$ — $5L$
z — $1L$

$z = 0,49\ mol/L$

Exercício 3 - Alternativa D

$0{,}3\ mol\ NaOH$ — $1000\ mL$
x — $40\ mL$

$\boxed{x = 0{,}012\ mol\ NaOH}$

$0{,}2\ mol\ KOH$ — $1000\ mL$
y — $60\ mL$

$\boxed{y = 0{,}012\ mol\ KOH}$

$NaOH_{(aq)} \rightarrow Na^+_{(aq)} + OH^-_{(aq)}$
$0{,}012\ mol \quad 0{,}012\ mol \quad 0{,}012\ mol$

$KOH_{(aq)} \rightarrow K^+_{(aq)} + OH^-_{(aq)}$
$0{,}012\ mol \quad 0{,}012\ mol \quad 0{,}012\ mol$

$0{,}012\ mol\ Na^+$ — $100\ mL$
z — $1000\ mL$

$\boxed{z = 0{,}12\ mol/L\ Na^+}$

$0{,}024\ mol\ OH^-$ — $100\ mL$
w — $1000\ mL$

$\boxed{w = 0{,}24\ mol/L\ OH^-}$

Exercício 4 - Alternativa E

$0{,}5\ mol\ NaOH$ — $1000\ mL$
x — $400\ mL$

$\boxed{x = 0{,}2\ mol\ NaOH}$

$40\ g\ NaOH$ — $1\ mol$
y — $0{,}2\ mol$

$\boxed{y = 8\ g\ NaOH}$

concentrada:

$80\ g\ NaOH$ — $1000\ mL$
$8\ g\ NaOH$ — z

$\boxed{z = 100\ mL\ da\ sol.\ conc.}$

volume final = $400\ mL$
volume conc. = $100\ mL$

$\boxed{volume\ H_2O = 300\ mL}$

Exercício 5 - Alternativa C

Solução 1: $0{,}8\,g \text{ — } 1000\,mL$ e $x \text{ — } 25\,mL$ $\Rightarrow$ $x = 0{,}032\,g$

Solução 2: $0{,}032\,g \text{ — } 100\,mL$ e $y \text{ — } 1000\,mL$ $\Rightarrow$ $y = 0{,}32\,g/L$ sol. 2

$0{,}032\,g \text{ — } 100\,mL$ e $z \text{ — } 5\,mL$ $\Rightarrow$ $z = 6 \cdot 10^{-4}\,g$

Solução 3: $6 \cdot 10^{-4}\,g \text{ — } 100\,mL$ e $w \text{ — } 1000\,mL$ $\Rightarrow$ $w = 0{,}006\,g/L$ sol. 3

Exercício 6

$$8CO_{2(g)} + 25H_{2(g)} \longrightarrow C_8H_{18(\ell)} + 16H_2O_{(\ell)}$$

$$\text{razão} = \frac{CO_2}{H_2} = \frac{8}{25}$$

O

$$8CO_{2(g)} + 25H_{2(g)} \longrightarrow C_8H_{18(\ell)} + 16H_2O_{(\ell)}$$

$8.(-394)$

$-3152\,KJ$

$-209 + 16.(-286)$

$-209 - 4576$

$-4785\,KJ$

$\Delta H = \text{Produto} - \text{Reagente}$

$\Delta H = -4785 - (-3152)$

$\Delta H = -1633\,KJ$

Exercício 7

A) A reação 3 libera calor, pois sua variação de entalpia é negativa, ou seja, a entalpia dos produtos é menor do que a dos reagentes.

B) $-286\,kJ$ — $1\,mol\ H_2O$
x — $4\,mol\ H_2O$

$x = -1144\,kJ$

C)

inv. $Cr_2O_{3(s)} + N_{2(g)} + 4H_2O_{(l)} \longrightarrow (NH_4)_2Cr_2O_{7(s)}$ $\Delta H = +476\,kJ$

ok $2Cr_{(s)} + \frac{3}{2}O_{2(g)} \longrightarrow Cr_2O_{3(s)}$ $\Delta H = -1128\,kJ$

x4 $4H_{2(g)} + 2O_{2(g)} \longrightarrow 4H_2O_{(l)}$ $\Delta H = -1144\,kJ$

$2Cr_{(s)} + 4H_{2(g)} + N_{2(g)} + \frac{7}{2}O_{2(g)} \longrightarrow (NH_4)_2Cr_2O_{7(s)}$ $\Delta H = -1796\,kJ$

Exercício 8 - Alternativa B

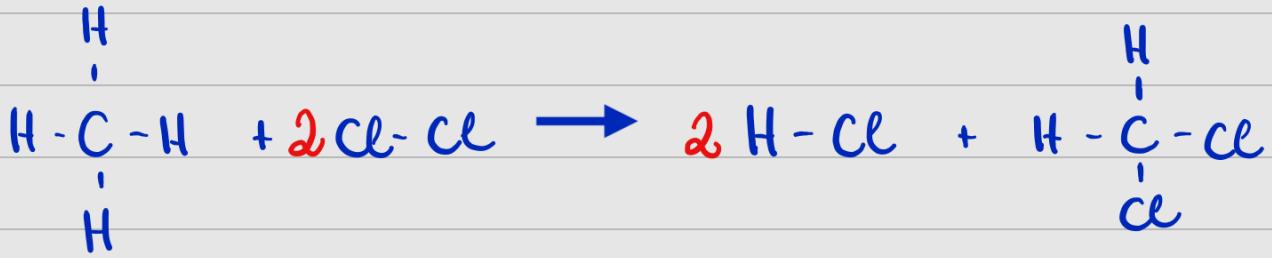
2x $2C_{(grafite)} + 2O_{2(g)} \longrightarrow 2CO_{2(g)}$ $\Delta H = -787\,kJ$

3x $3H_{2(g)} + \frac{3}{2}O_{2(g)} \longrightarrow 3H_2O_{(l)}$ $\Delta H = -857{,}4\,kJ$

inv. $2CO_{2(g)} + 3H_2O_{(l)} \longrightarrow C_2H_{6(g)} + \frac{7}{2}O_{2(g)}$ $\Delta H = +1558{,}3\,kJ$

$2C_{(grafite)} + 3H_{2(g)} \longrightarrow C_2H_{6(g)}$ $\Delta H = -86{,}1\,kJ$

Exercício 9 - Alternativa A



$$2.(C-H) + 2(Cl-Cl) \longrightarrow 2.(H-Cl) + 2(C-Cl)$$

$$2.430 + 2.240 \qquad 2.430 + 2.330$$

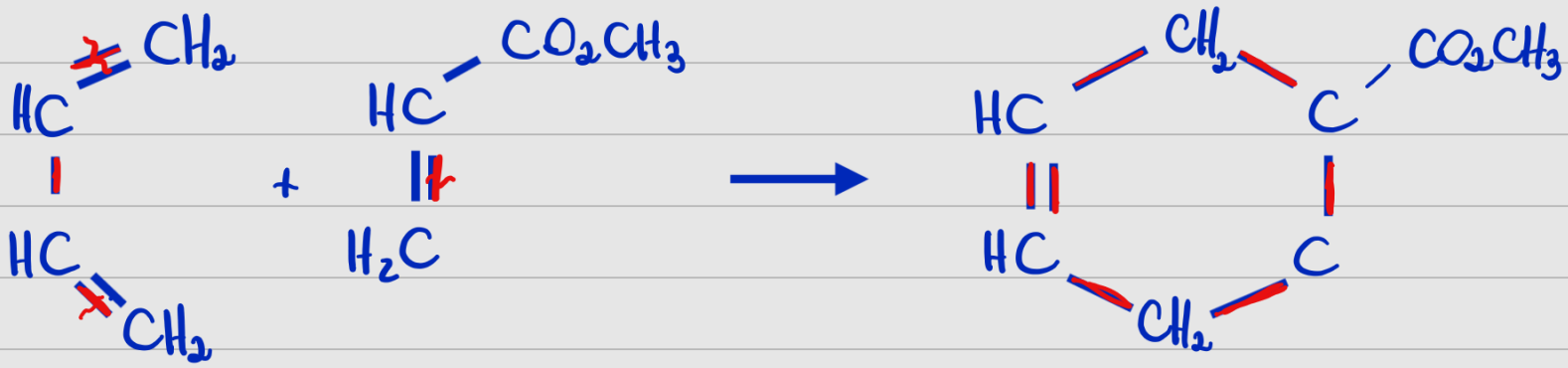
$$820 + 480 \qquad 860 + 660$$

$$+1300\ kJ \qquad -1520\ kJ$$

$$\Delta H = +1300 - 1520$$

$$\boxed{\Delta H = -220\ kJ}$$

Exercício 10 - Alternativa A



$$3(C=C) + 1(C-C) \qquad 1.(C=C) + 5(C-C)$$

$$3.612 + 348 \qquad 612 + 5.348$$

$$+1836\ kJ + 348 \qquad 612 + 1740$$

$$+2184\ kJ \qquad -2352\ kJ$$

$$\Delta H = +2184 - 2352$$

$$\boxed{\Delta H = -168\ kJ/mol}$$

Exercício 11 - Alternativa C

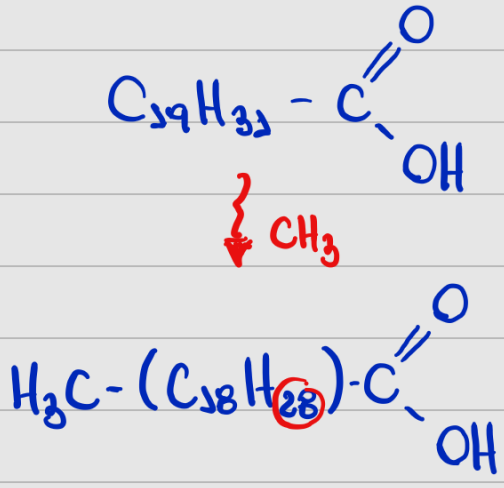
↠ triésteres de glicerina (triglicerídeos) e ácidos graxos **saturados** são predominantes nos **gorduras**.

↠ triésteres de glicerina (triglicerídeos) e ácidos graxos **insaturados** são predominantes nos **óleos**.

ácido graxo + glicerol ⟶ triglicerídeo + água
(lipídeos)



Exercício 12 - Alternativa A



simples ↠ $CH_2 - CH_2 \cdots$

C_nH_{2n}

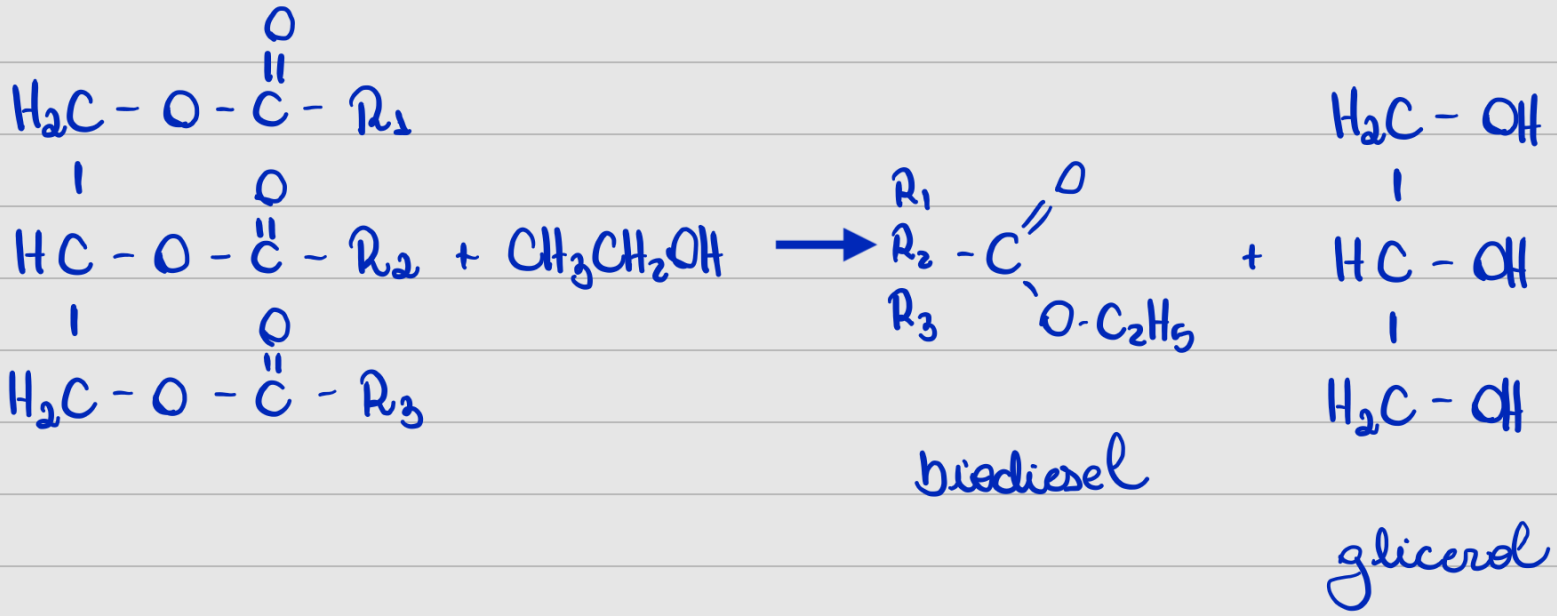
cada dupla ↠ - 2 Hidrogênios

$C_{18}H_{36}$ ⟶ $C_{18}H_{28}$

8 hidrogênios

4 duplas ligações

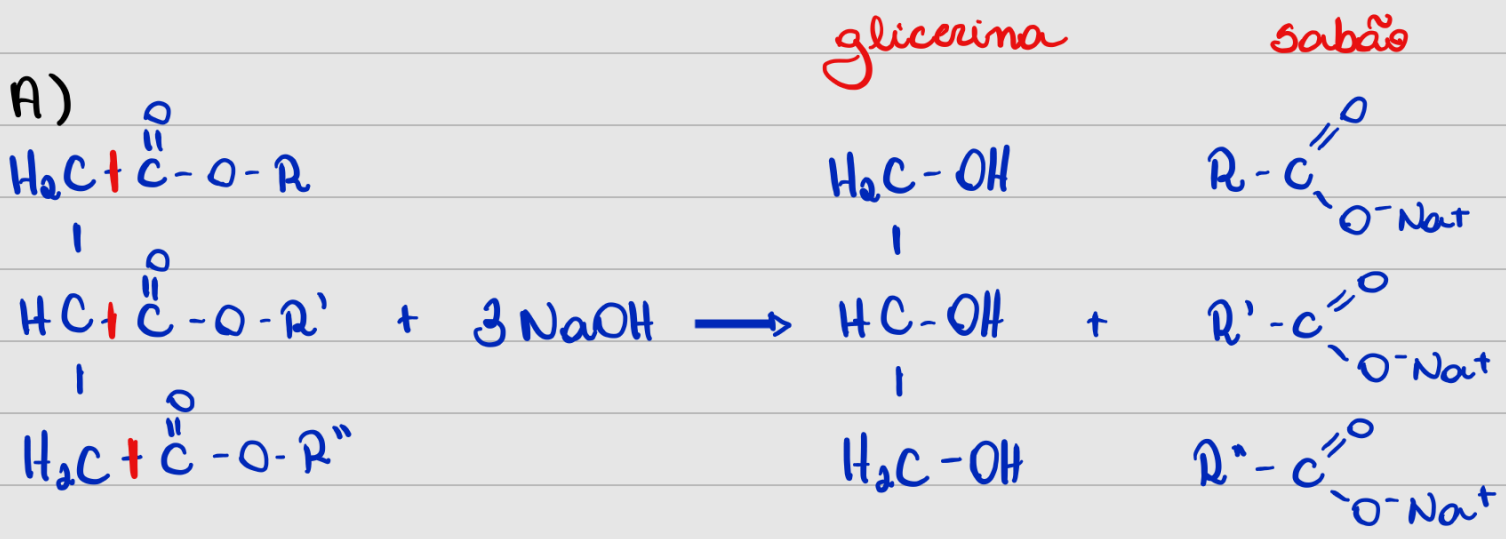
Exercício 13 - Alternativa A



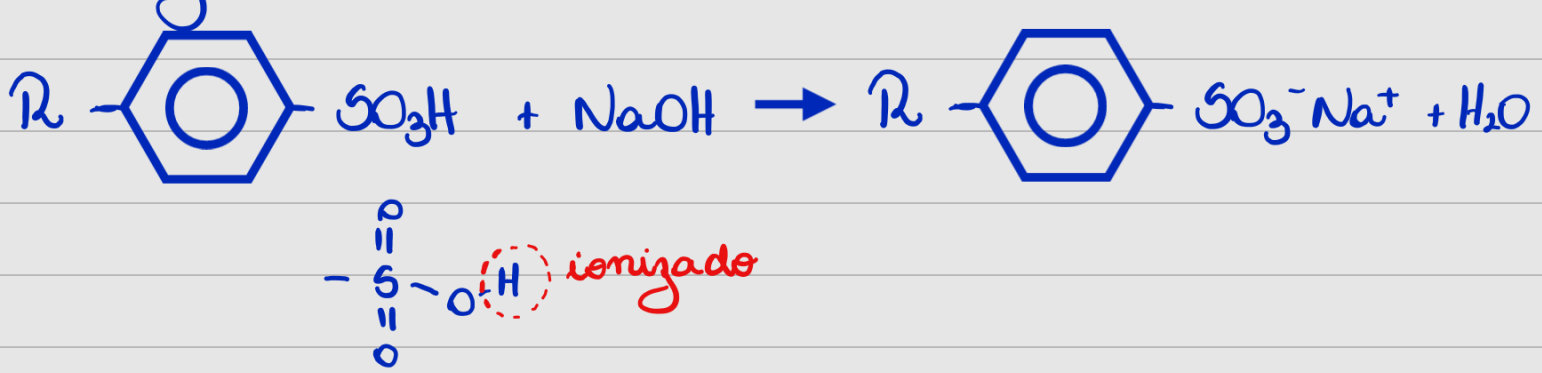
Exercício 14 - Alternativa A

Exercício 15

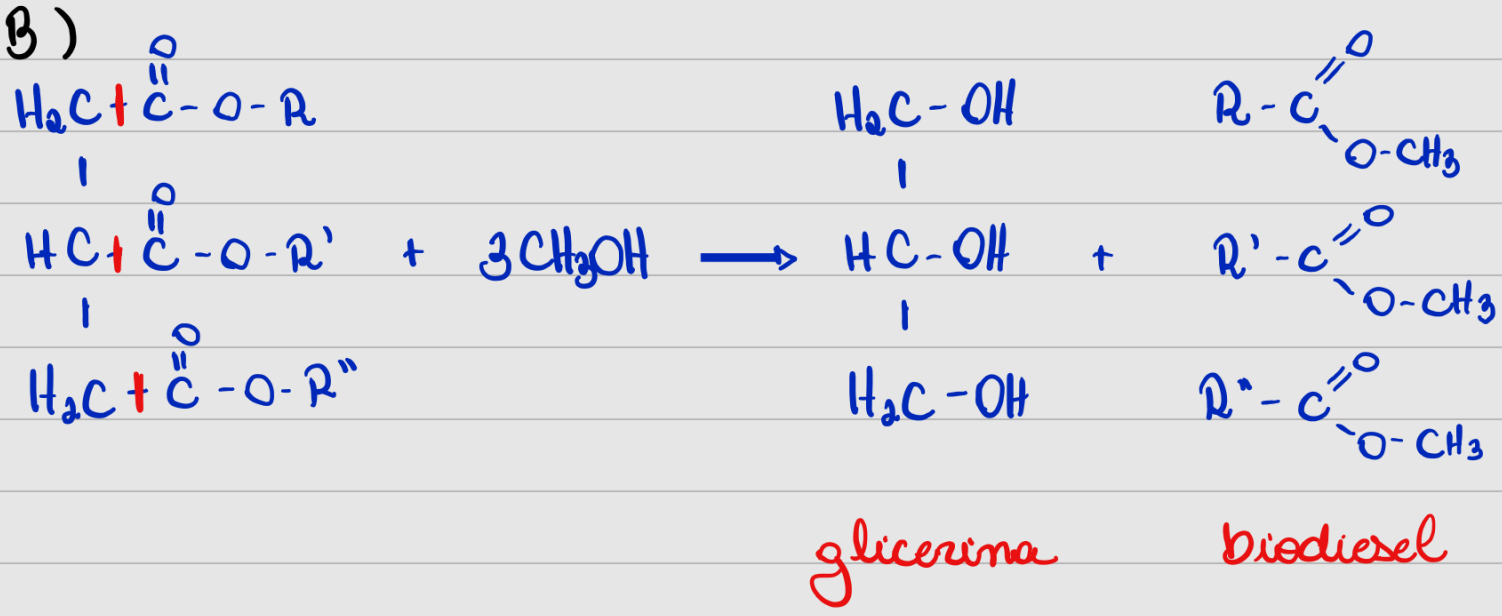
A)



* Detergente:



B)



glicerina

biodiesel